



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **Soft Switching Converter Technologies**
Course Title : **EEE G554**
Instructor-in-Charge : **Rajneesh Kumar**

1. Course Description:

This course covers analysis of series and parallel resonant converters, load dependent properties of resonant converters with ZVS/ZCS boundary. Operation of these converters for Zero-current Switching and Zero-Voltage switching is discussed in detail. The concept of resonance is extended to analyze the soft switching mechanism of semiconductor devices Diode, MOSFET and IGBT to reduce the switching loss and possible mitigation of EMI. The zero-current-switching and the zero-voltage-switching quasi resonant switch cell is discussed for full bridge converter. Design example of zero-current-switching and zero-voltage switching class E-inverter is also covered in this course.

2. Scope and Objective of the Course:

- Understand the concept of resonant tank network*
- Analyze the dependence of ZVS/ZCS boundary on load resistance*
- Design ZVS/ZCS switch cell for DC-DC converter*
- Analysis of switching loss and EMI under soft switching*

3. Text Books:

1. **Fundamentals of Power Electronics** by Robert W. Erickson, Dragan Maksimović, 2nd edition, 2001 Kluwer Publication.

4. Reference Books:

1. **Resonant Power Converters** by Marian K. Kazimierczuk and Dariusz Czarkowski, 2nd edition, Wiley

5. Course Plan:

Lecture No.	Learning objectives	Topics to be covered	Chapter in the Text Book
1	Introduction	Introduction	
2-3	Basic Resonant Converter Topologies	Series, parallel, series-parallel resonant DC-DC converters, half and full bridge topologies, analysis and design	19
4-6	Analysis of Resonant Converters	Sinusoidal analysis of resonant converters, analysis and design	19.1-2
7-9	Soft Switching	Operation of the Full Bridge Below Resonance: ZCS, ZVS	19.3
10-12	Load-Dependent Properties of Resonant Converters	Inverter output characteristics, Dependence of Transistor Current on Load, Dependence of the ZVS/ZCS Boundary on Load Resistance, Example of LLC tank, Results for Basic Tank Networks	19.4



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

13-15	Exact Characteristics of the Series and Parallel Resonant Converters	Exact Characteristics of the Series and Parallel Resonant Converters	19.5
16-17	Soft-Switching Mechanisms of Semiconductor Devices	Diode Switching, MOSFET Switching, IGBT Switching	20.1
18-20	Zero current and Zero voltage switching quasi resonant converters	Zero-Current Switching Quasi-Resonant Switch Cell, Resonant Switch Topologies,	20.2-3
21-23	Soft Switching in PWM Converters	Zero-Voltage Transition Full-Bridge Converter, Auxiliary Switch Approach, Auxiliary Resonant Commutated Pole	20.4
24-25	Multi Resonant Converters	ZVS and ZCS multi-resonant dc-dc Converters	22.10-22.11 (Ref)
26-30	ZVS Resonant Inverters	Principle of operations, Analysis, Parameters at $D = 0.5$, Efficiency, Matching resonant circuit	12 (Ref)
31-35	ZCS Resonant Inverters	Principle of operations, Analysis, Power relationships, Element values of load network	13 (Ref)
36-38	EMI Suppression	Concept of EMI, conducted and radiated EMI, common mode voltage, input filter design, common mode choke, snubber circuits	(Technical Papers)
39-40	Passive components at high frequencies	Effect of high frequency on passive components and transformers, frequency domain analysis (bode plot)	(Technical Papers)

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Maximum Marks	Date & Time	Nature of Component
Quiz (2 No)	30 min each	10	20M	TBA	OB
Simulation Assignments	TBA	20	40M	TBA	OB
Literature Survey and Seminar	TBA	10	20M	TBA	OB
Mid-Sem	90 min	25	50M	TBA	CB
Comprehensive	180 min	35	70M	TBA	OB



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

7. Chamber Consultation Hour: TBA

8. Notices: on NALANDA

9. Make-up Policy: As per the institute guidelines

10. Attendance Policy: NA

11. Minimum passing Marks criteria: NA

12. Note (if any): NA

Rajneesh Kumar

Instructor-in-charge
Course No. EEE G554